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CAUSES AND CONTROL OF CRACKING IN UNREINFORCED MASS CONCRETE

by

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SYNOPSIS

The most common cause of cracking is restrained contraction. Principal causes of contraction are loss of moisture (drying shrinkage) and a decrease in temperature. Control of cracking by reducing drying shrinkage, change in temperature, and by modifying the properties of concrete, are outlined.

KEYWORDS

Aggregates; coefficient of thermal expansion; cement (type); contraction; compressive strength; concretes (plain, mass); concrete cooling (pre- and post-cooling); cracking (fracturing); heat of hydration; modulus of elasticity; pozzolans; restraint; relaxation (stress); shrinkage (autogenous, drying); temperature change (interior, surface); tensile properties (strain, stress, strength).

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INTRODUCTION

Concrete is in some respects a brittle material. Its compressive strength is roughly ten times its tensile strength; therefore, plain concrete serves well where resistance to crushing is needed. It can be made to have a predetermined and dependable compressive strength. On the otherhand, plain concrete cannot be depended upon to support tensile stress because too often it cracks due to causes other than applied loads.

The cracking of concrete is undesirable for many reasons. Besides being unsightly, cracks may reduce the safety of a structure by preventing monolithic action and, in some cases, they may affect such properties as durability and water-tightness.

The purpose of this paper is to define the factors, other than applied loads, which cause cracking and to suggest measures which can be taken to minimize the cracking of plain concrete. The suggested measures are of two types. First, factors which produce tensile stress can be modified. Here the emphasis is on control of temperature changes and drying shrinkage [1]. Second, the resistance of the concrete against cracking can be improved. Here the emphasis is mainly on the properties of the concrete, such as tensile strain capacity.

The tendency of concrete to contract is caused primarily by a loss of moisture or a decrease in temperature. Autogenous shrinkage [2] is an infrequent cause of contraction, and therefore, will not be discussed in this paper. The effect of temperature changes will be considered first, while cracking due to shrinkage will be deferred until later.

In what follows it should be borne in mind that the measures suggested for the prevention of cracks are not always necessary. The suggested measures apply mainly to structures where one or more of the following conditions prevail: (a) large dimensions of concrete, (b) high stresses, (c) severe climate, (d) aggregates with unfavorable properties. In all cases, preliminary tests and studies should be made which may show that only a well designed mix and careful construction may suffice without the need for special requirements.

CRACKING DUE TO TEMPERATURE CHANGE

In order to determine whether or not cracking will occur due to temperature change, the first step is to estimate the tensile stress or strain which develops. It is well known that cement liberates heat when it hydrates and that concrete is more or less plastic during the rise in temperature which results. After a maximum temperature has been reached [3,4] and cooling occurs, the concrete is stronger and more nearly elastic. If there is any restraint against free contraction during cooling, tensile stress develops, which may be enough to cause cracking.

For practical purposes, a simple method can be used for computing the approximate tensile stress due to internal cooling. It has been found that at the time of maximum internal temperature, the stress is near enough to zero that it can be assumed to be zero. As a first step, one can assume that there is no relaxation due to the slowness of stress development. The tensile stress is then merely the product of four quantities, as follows: (1) temperature drop in degrees, (2) coefficient of thermal expansion in millionths per degree (3) the average modulus of elasticity during the temperature drop in psi, kgf/cm^2, or MPa, and (4) the degree of restraint as a fractional part of full restraint. Since the temperature drop and resulting stress never occur instantaneously, a correction needs to be applied to allow for relaxation. There is now an easy way of determining the relaxation of stress due to slow loading, which is described in reference [5]. The method makes use of relaxation coefficients as derived from creep data. Suffice it to say here that it is beneficial to have the temperature drop occur as slowly as possible and to reduce the restraint against contraction whenever practicable.

It has been shown that four quantities, plus the relaxation, are needed to determine the tensile stress in the concrete. Opposing the stress is another important factor which must be known to determine if cracking will occur; this is the tensile strength.

Control of Temperature Change

Temperature changes are caused by either exposure to weather conditions externally or heat generation of the cement internally [6]. In regions where weather fluctuations are very large, it is often practicable to cover exposed surfaces with thermal insulation to prevent minor surface cracking during cold weather. Nearly all cracks originate at exposed surfaces, so the prevention of what appear to be minor surface cracks may amount to preventing major cracks.

The most serious temperature change is the drop which occurs after a maximum has been reached in the interior. Although in structures such as dams this drop in temperature may be slow, a steady state must finally be reached which depends upon the boundary conditions. In a typical case that occurs in dams, the final temperature will vary from water temperature near the upstream face to the yearly average ambient temperature near the downstream face. Thus, it is essential that the maximum internal temperature be kept as low as practicable so the ultimate drop in temperature will not cause the concrete to crack.

There are many ways to keep the interior temperature from rising too high. Some of them, in approximate order of effectiveness, are as follows:

(a) precooling of the fresh concrete;

(b) postcooling by the passage of cold water through pipes embedded in the concrete;

(c) replacing part of the cement with a good pozzolan;

(d) casting the concrete in thinner lifts;

(e) reducing the cement content to only what is necessary for strength;

(f) using a cement composition with low heat of hydration; and

(g) extending the period of time between placement of successive lifts.

Precooling of the fresh concrete -- When concrete is cast at a reduced temperature, not only is the subsequent maximum reduced but there are other advantages as well. At lower temperatures, the concrete is more fluid, so the water and cement content can be reduced as much as five percent. Also, at the lower temperature the cement hydrates more slowly so there is less thermal stress due to differences within each lift. When the release of heat is more gradual, the underlying concrete or foundation participates more in the temperature change and thus the restraint is reduced. When precooling of the concrete is adopted, it is often specified that the temperature of the fresh concrete shall not exceed 45°F (7°C). However, cooling to such a low temperature is expensive, so it is sometimes considered sufficient to cool less than this. The amount of cooling which is necessary can be estimated by combining past experience with a study of job conditions and concrete properties.

Pipe cooling by the use of embedded pipes -- The maximum temperature of interior concrete can be kept low by pumping cold water through embedded pipes, but this measure is seldom justified for temperature reduction alone. In a very thick dam, pipe cooling is almost mandatory if the dam is to be brought to a stable condition before being placed in service. This becomes clear when the alternative of no pipe cooling is considered. In this case, if the entire dam was cast as a monolith, cracks are likely to occur and these cracks will continue to open for scores of years, thus tending to prevent monolithic action. If joints were to be provided to allow for the contraction, they would not open enough to allow them to be grouted for many years and, after grouting, they would continue to open. The best solution, even though it is expensive, is to make formed joints, provide for cooling of the concrete to a final state after the heat evolution is substantially complete, and then to grout the joints.

Table 1 shows the approximate time required for 90 percent of the temperature drop to occur after a maximum has been reached and heat generation has ceased [7]. Objects being cooled are concrete walls of varying thicknesses, cooling from opposite faces only. The fastest cooling is for concrete containing quartzite aggregate, diffusivity 1.3 ft^2/day (0.12 m^2/day), and the slowest is for concrete containing basalt aggregate, diffusivity 0.7 ft^2/day (0.06 m^2/day). Average concrete has a diffusivity of 0.9 ft^2/day (0.08 m^2/day).

Replacing part of the cement by a good pozzolan -- Pozzolan is a finely divided siliceous material which can react with the calcium hydroxide liberated when the cement hydrates. A good pozzolan can be used to replace up to 30 percent of the cement by absolute volume without any sacrifice of long-term strength. A 30 percent replacement of cement by a good pozzolan will reduce the heat generation by about 15 percent.

There are other benefits of using a good pozzolan. It can reduce or prevent expansion due to alkali-silica reaction in cases where the aggregate is reactive. It can be used to supply needed fines and to prevent bleeding in very lean mixes. Finally, there is usually a savings in cost when the pozzolan is used as a replacement for part of the cement.

Casting the concrete in thinner lifts -- For concrete which is not precooled, there are several benefits from casting in shallow lifts. More heat escapes to the surroundings, so the maximum internal temperature is reduced. Smaller differences in temperature occur within each lift and this reduces the stresses due to nonuniform temperature. Also, thinner lifts encourage better bonding of new concrete to old. It should be emphasized that these benefits make up only partially for the larger benefits which would have accrued if the concrete had been adequately precooled.

When the concrete is precooled to about 45°F (7°C) and the cement content is low, there is very little benefit from casting in shallow lifts. In general, heat absorbed from the surroundings during the early stages of cement hydration will partially offset the heat loss thereafter. When the cement content is high and the ambient temperature is low, there is some benefit from shallow lifts, even for precooled concrete.

Reducing the cement content -- The internal heating of concrete comes from the hydration of the cement, so the lower the cement content, the lower the heat and the lower the temperature rise (see Fig. 1). It has been observed in a number of mass concrete structures that the only concrete completely free from cracks was that containing the least amount of cement per cubic meter. This cannot be explained solely on the basis of temperature differences, so lean concrete must have better properties for resisting cracking. Possible explanations are the lower modulus of elasticity and the lesser tendency toward drying shrinkage. When there are surface cracks due to drying shrinkage, concentration of stress occurs at crack tips such that cracks can propagate with only half as much applied tensile stress as would be required otherwise.

Using a cement composition with low heat of hydration -- By controlling the chemical composition of the cement, both the rate and total amount of temperature rise can be reduced to advantage. The goal is to use a composition which gives the lowest heat of hydration per unit of compressive strength of the concrete. The worst constituent in this regard is the tricalcium aluminate, so limiting its percentage to 7 percent or less is helpful. Also, limiting the tricalcium silicate, with a corresponding increase in dicalcium silicate, is usually beneficial, an exception being when the cement content is so low that a fairly high content of tricalcium silicate is needed for early strength. The requirements for most applications can be met by specifying either Type IV (low heat) or Type II (moderate heat) cement rather than Type I, which is a high-heat cement.

Controlling the finensss of the cement -- The fineness of cement affects mainly the rate at which heat is liberated, while having almost no effect on the ultimate heat liberation. Careful tests have shown that the finer the cement, the higher the strength of concrete at all ages, and not only at the early ages. There are sometimes cases where the use of a very fine cement is advantageous. For example, if the concrete is precooled to under 45°F (7°C) and the cement content is very low, the concrete will hydrate slowly for several days even when finely ground cement is used. In such cases, less of the finer cement needs to be used to obtain the desired strength, and the maximum interior temperature will be less. However, if the cement content is not very low, even with the low placing temperature, the concrete will soon by hydrating faster than desired. The rise in temperature acts as an accelerator and the consequent fast release of heat can cause undesirable thermal gradients and stresses.

When concrete in a large mass hydrates too quickly, not only is there danger of serious thermal stress and cracking, but bonding at work joints becomes a problem. When the concrete in a new lift warms too rapidly due to heat of hydration, the new concrete tends to expand over the underlying concrete just at the time when there should be no relative movement. Then good bonding is often prevented. Some fifty years ago, a large percentage of work joints in concrete dams leaked badly because of a bad combination of high-heat cement, rich mixes and no precooling. When precooling of the concrete is uneconomical, consideration should be given to using shallow lifts, slow hardening cement, and only enough cement to provide adequate strength. Even in severe climates the interior concrete could have a low cement content if a layer of richer concrete is used at exposed faces.

CRACKING DUE TO DRYING SHRINKAGE

Cracking of concrete due to drying shrinkage is one of the most common and serious problems encountered in concrete structures. As in the case of cracking due to decrease in temperature discussed earlier, the cause of cracking due to drying shrinkage is the combination of the tendency to contract and the restraint which prevents it from doing so. When the resulting tensile stress reaches the tensile strength of the concrete, it will crack.

The tensile stress development which may lead to cracking involves the following factors: (1) rate and amount of drying shrinkage, (2) modulus of elasticity, (3) degree of restraint, and (4) creep or relaxation of tensile stress during shrinkage. Factors influencing drying shrinkage are discussed in this section of the paper. Modulus of elasticity and the relaxation of concrete are properties influencing its strain capacity, which are covered in the subsequent sections of this paper.

The restraint of concrete in a structure comes from several sources. For example, some degree of restraint is provided by the foundation or another part of the structure. A very common type of restraint is developed by the difference in shrinkage at the surface and in the interior of a concrete member, especially at early ages of exposure to drying. Since the drying shrinkage is largest at the exposed surface, the interior concrete restrains the shrinkage at the surface, thus developing tensile stresses. This may cause surface cracks, which do not penetrate deep into the concrete. However, as drying of the concrete continues, the surface cracks will frequently penetrate deeper into the concrete and become major structural cracks.

In the case of a mass concrete structure or large structural elements, the problem of surface cracking would be aggravated if drying and cooling of the surface should occur simultaneously, resulting in high tensile stress development. For this reason it is important to protect the surfaces of large structural elements from drying during the concrete's cooling period. This can be accomplished for example by continued curing or by sealing the surfaces to minimize moisture loss.

Factors Influencing Drying Shrinkage

As shown by Carlson [8], the drying shrinkage of concrete is influenced by many factors, some of which have a significant effect while others are of minor importance. The major factors influencing shrinkage include the (a) water content and mix proportions, (b) composition of cement, and (c) type and maximum size of aggregate. The rate and amount of moisture loss, or drying shrinkage, is greatly influenced by the size and shape of the concrete member, wind and the relative humidity of the environment. Some of these factors influencing drying shrinkage are discussed here in more detail.

Effect of water content and mix proportions -- It is mainly the loss of water from the cement paste which causes the contraction of concrete on drying, and thus the water content is a major factor influencing shrinkage. An increase in water content of a concrete mix will not only increase its water-cement ratio but it will also reduce the volume of the aggregate, thus further increasing the shrinkage of the concrete. For example, a 15 percent increase in the water content of a structural concrete mix can increase its drying shrinkage by about one-third.

In proportioning a concrete mix the potential shrinkage can be reduced by the use of a larger aggregate size and lower water content. Concrete mixes containing small size aggregate and high sand content, as frequently used for pump placement, will have high drying shrinkage. At a given water content, changes in cement content will have only a small influence on shrinkage, with richer mixes having higher shrinkage since they have a lower aggregate content.

Effect of cement -- According to Carlson [8], the variations in concrete shrinkage for different portland cements cannot be explained satisfactorily by differences in their composition or fineness. This was confirmed by Blaine et al [9] whose test results show that it is not possible to say that a cement conforming to one of the standard ASTM types of portland cements will produce a greater or lower shrinkage than a cement meeting the requirements of one of the other types of cements. Their results on cement pastes showed a wide distribution of shrinkage values, especially for the Type I cement.

The chemical composition of a cement, however, does have an important influence on shrinkage. Of the several factors which can influence the magnitude of drying shrinkage, the important ones are the sulfate (SO_3), C_3A, and alkali (Na_2O and K_2O) content of the cement. The SO_3 is added in the form of gypsum to control setting during grinding of the cement clinker. Tests by Lerch [10] have shown that the proportion of gypsum in the cement has a major effect on shrinkage. Lower shrinkage characteristics of cement pastes are associated with lower C_3A/SO_3 ratios and lower Na_2O and K_2O contents.

Influence of aggregate -- Other important factors influencing drying shrinkage of concrete are the size and rigidity of the aggregate, since the aggregate provides an internal restraint which significantly reduces the potential contraction of the paste. Depending on the type and amount of aggregate used, the drying shrinkage of a concrete will be only about one-fourth to one-sixth of the shrinkage of cement paste. It is the stiffness or modulus of elasticity of the aggregate which influences its ability to restrain shrinkage. The less compressible the aggregate, the lower the shrinkage of the concrete. The large influence of type of aggregate on drying shrinkage was shown by Troxell et al [11]. For example, the one-year shrinkage of concretes of fixed proportions but made with different aggregates are given in Table 2.

Low shrinkage concretes may be produced by using quartz, dolomite, granite, feldspar, and some limestones, whereas the use of graywacke, sandstone, slates, hornblende, and some basalts may produce high shrinkage concretes. The modulus of elasticity of certain aggregates, such as basalt, limestone, and dolomite, can vary over a wide range. Hence, their effectiveness in restraining drying shrinkage will also vary.

In addition to the compressibility of an aggregate, which is one of the main factors influencing shrinkage of the concrete, the aggregate itself may exhibit a high contraction upon drying. This is the case for sandstone or graywacke, both having high absorption capacities. In general, the use of an aggregate of high absorption and low modulus of elasticity will produce high shrinkage concretes. This, however, is not true for the lightweight aggregates used for structural concretes, such as expanded shales, clays, or slates. Some of these lightweight aggregates, which have high absorptions, produce concrete having shrinkage characteristics which are similar to or even lower than those of corresponding normal weight aggregate concretes [12].

The maximum size of aggregate used also has a significant influence on shrinkage, principally as it has an effect on the water content of a concrete mix. A larger aggregate size will not only reduce the water content, and thus shrinkage, but is more effective in restraining the shrinkage of the paste.

Influence of size of member -- The rate and magnitude of drying shrinkage of concrete in a structure will be substantially smaller than the shrinkage of the same concrete when determined by test on small laboratory specimens. Thus, the size and shape of a concrete member has a significant influence on shrinkage. As shown by Carlson [13], at a 50 percent relative humidity, drying will penetrate only about 3 inches (75 mm) from the surface of a large concrete member during the first 30 days, and it will take 10 years to reach a depth of 2 feet (0.6 m). Results of an extensive investigation by Hansen and Mattock [14] on the influence of size and shape of a member show that ratio of volume to its surface area is a suitable parameter when estimating its shrinkage or creep deformations. They found that both the rate and the projected final values of shrinkage and creep decrease as the member becomes larger.

Effect of humidity of the environment -- The higher the relative humidity of the environment, the lower the ultimate shrinkage of the concrete. Since concrete loses its moisture extremely slowly, but easily absorbs moisture from the atmosphere, it can be assumed that the concrete in the structure is for all practical purposes exposed to a humidity equal to the average annual relative humidity of the atmosphere. If this average relative humidity is, for example, 70 percent, then, as reported by Troxell et al [11], the drying shrinkage would be about one-third lower than if the concrete was exposed to a 50 percent relative humidity.

Control of Cracking Due to Drying Shrinkage

The cracking of concrete in a structure can be controlled and minimized by (a) reducing its shrinkage characteristics, as discussed above; (b) using control joints which will force cracks to form in the grooves of these joints, thus relieving the tensile stress and preventing the formation of additional cracks; and, (c) coating or sealing the concrete surface which will minimize drying and thus reduce the tensile stress development causing cracking.

Increasing the strain capacity of a concrete is another way to reduce cracking. The properties of concrete which influence its strain capacity are the subject covered in the next section.

PROPERTIES OF CONCRETE AFFECTING CRACK RESISTANCE

The principal properties of concrete affecting its tendency to crack are (1) coefficient of thermal expansion, (2) tensile strain capacity, (3) tensile strength, (4) modulus of elasticity, (5) heat of hydration, (6) drying shrinkage, and (7) autogenous shrinkage. The heat of hydration and drying shrinkage have already been discussed. Autogenous shrinkage will be omitted because it is usually negligible, although it should not be assumed so without test.

Much information has been published about the physical properties of concrete. Specific information about those properties which affect cracking may be found in papers by Houghton [15,16]. Only a brief review of such properties will be offered here, with suggestions as to how they might be modified to improve the resistance against cracking.

Coefficient of Thermal Expansion

When the concrete is wholly restrained, the unit tensile strain is simply the product of the coefficient of thermal expansion and the temperature drop in degrees. Such complete restraint is approximated when a new section of concrete is cast on a rigid foundation or on old concrete. Similarly, nearly full restraint occurs also at exposed surfaces of mass concrete subjected to a rapid drop in ambient temperature. Under this latter condition, contraction of the concrete at the surface is resisted by the internal mass which may be increasing in temperature or at least not decreasing. In these cases, where the restraint can be assumed to be 100 percent, the strain can be computed simply as stated above. If the temperature distribution could then in some way be maintained constant, the strain would remain constant but the corresponding stress would relax and gradually become less. A method for converting strain into stress and taking account of relaxation as time goes on is described in reference [5].

The magnitude of the coefficient of thermal expansion depends largely upon the mineral composition of the aggregate. Therefore, the only way it can be changed appreciably is to use a different aggregate. Low coefficients of thermal expansion are usually found when the aggregates are limestones or basalts, while high values are found with quartzitic aggregates. The range of values reported for concretes from various projects runs from a low of approximately 4 millionths per °F (7 millionths per °C) to a high of over 7 millionths per °F (13 millionths per °C). It is apparent that the choice of an aggregate of low thermal expansion would contribute favorably to crack prevention.

Tensile Strain Capacity

Houghton [15,16,17] has described the technique for measuring the strain capacity of concrete through the use of plain beams loaded at the third points. A simple method of estimating the approximate strain capacity has also been described by Houghton in the December 1976 issue of the ACI Journal [15]. In this method, the modulus of rupture as determined in the usual way is simply divided by the modulus of elasticity in compression to obtain the ultimate tensile strain for quick loading. The added strain capacity due to slow loading is determined by taking account of the creep. The method would be exact if the modulus of rupture were the true tensile strength and if the modulus of elasticity in compression were the true value of stress-strain ratio at failure and in tension. Because the modulus of rupture is some 20 to 40 percent higher than the true tensile strength and the modulus of elasticity in compression is about this much higher than the stress-strain ratio in tension, the approximate method gives nearly true values. Although the approximate methods are useful for preliminary planning, they should not be used to replace direct testing for important work.

A few data on tensile strain capacity have been extracted from one of Houghton's papers [15] and are tabulated in Table 3. It may be noted that the strain capacity of the neat cement paste is higher than for any of the concretes or mortar and that there is a consistent decrease in strain capacity with increasing size (and consequent amount) of aggregate. The data show, also, the effect of age; there being a consistent improvement in the strain capacity with increasing age, with one exception.

The advantage of crushed quartzite aggregate over natural quartzite with well-rounded particles is apparent in Table 3. An important trend not shown in Table 3 is the large increase in strain capacity which was observed when the tensile strain was allowed to develop slowly. The increase was especially large when the aggregate was crushed instead of natural. In this case, where the 28-day strain capacity was already large (106 millionths), the value for slow loading up to the age of 90 days was more than double, or 215 millionths. When the aggregate was natural quartzite made up of well-rounded particles, the improvement was less striking but nevertheless substantial. Here the increase from the 28-day quick load value to the 90-day slow load value was from 71 to 117 millionths. No quick load values at 90 days were available in these cases. The manufactured aggregate causes only a slight increase in the required cement content and should be investigated for projects where extensive crack control measures are necessary. Also, insulation to prevent rapid cooling of the surfaces of mass concrete subject to low ambient temperature would help prevent cracking.

The effect of richness of mix on strain capacity may be seen by comparing Mix Nos. 2 and 6, where the cement contents were 220 and 550 lbs. per cubic yard (131 and 550 kg/m^3), respectively. The richer mix showed a strain capacity which was approximately double that of the leaner mix at the early age of 7 days, but only about 1.5 times as great at 28 days. It was reasoned that the higher paste content and higher strength of the richer mix tended to increase the strain capacity, while the higher modulus of elasticity tended to decrease it. It is evident that the combined effects of higher paste content and higher strength are greater than the opposing effect of higher modulus of elasticity, where strain capacity is concerned.

In practice, the apparent benefits of higher cement content are not realized. The greater heat generation and drying shrinkage of the richer mix more than overbalance the better strain capacity, except in thin sections not exposed to drying. As an experiment on one construction project where extensive cracking was encountered, a layer of rich concrete was used to top out a massive lift of leaner concrete. Even though the rich concrete was purposely limited to a thin layer so as to minimize the effect of the greater heat generation, cracking occurred at the top surface as before. It was concluded that the rich mix was not beneficial even in the thin layer and that if the rich mix had been used throughout each massive lift, the cracking would have been intolerable.

Tensile Strength

Past practice has been to convert strain to stress before calculating safety against cracking. The reason for this is that the tensile strength remains substantially constant at each age, regardless of the rate of stress development (exclusive of impulsive loading), while the strain to failure increases greatly with slow loading. However, thanks to the research of Houghton and others, it is now equally convenient to use either ultimate strain or ultimate stress in analyzing cracking tendencies. It is only necessary to take account of creep to determine how much additional strain the concrete can withstand under slow loading. This may be advantageous because of the difficulty of determining the true tensile strength.

The simplest method of measuring tensile strength is by direct pull on cylindrical specimens. However, this method is cumbersome and the results usually show considerable scatter. Therefore, it is often preferable to use either the tensile splitting test as detailed in ASTM C496 or beams from which can be derived the tensile strength as the modulus of rupture. The modulus of rupture, although it is the theoretical tensile strength, is higher than the true tensile strength for two reasons. Only a small part of the beam is under the maximum stress so the chance of an early failure is small. Also, the stress distribution across the beam is nonlinear, which makes the computed stress (modulus of rupture) too high. Therefore, a conservative method of determining tensile strength is to determine the modulus of rupture and then take 60 percent of this value or take some other percentage which test data will support.

Modulus of Elasticity

This property of concrete is important because it has a very large effect on both the strain capacity and the creep or relaxation. The lower the modulus of elasticity, the more strain the concrete can withstand and the less the likelihood of cracking. The modulus of the concrete is governed largely by the modulus of the aggregate, although the cement content is also an important factor. Since the cement content is usually that which is necessary for strength, the only practicable way the modulus of elasticity of the concrete can be varied is to choose a different aggregate. For crack prevention in plain mass concrete the aggregate with lower modulus of elasticity will be superior.

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TABLE 1. TIME REQUIRED TO ATTAIN A 90 PERCENT TEMPERATURE DROP OF MASS CONCRETE

Thermal Diffusivity	Time Required at Various Thicknesses					
	10 ft (3 m)	20 ft (6 m)	50 ft (15 m)	100 ft (30 m)	200 ft (60 m)	400 ft (120 m)
0.7 ft^2/day (0.06 m^2/day)	41-d	166-d	2.8-yr	11-yr	45-yr	181-yr
0.9 ft^2/day (0.08 m^2/day)	32-d	128-d	2.2-yr	9-yr	35-yr	141-yr
1.3 ft^2/day (0.12 m^2/day)	22-d	89-d	1.5-yr	6-yr	24-yr	98-yr

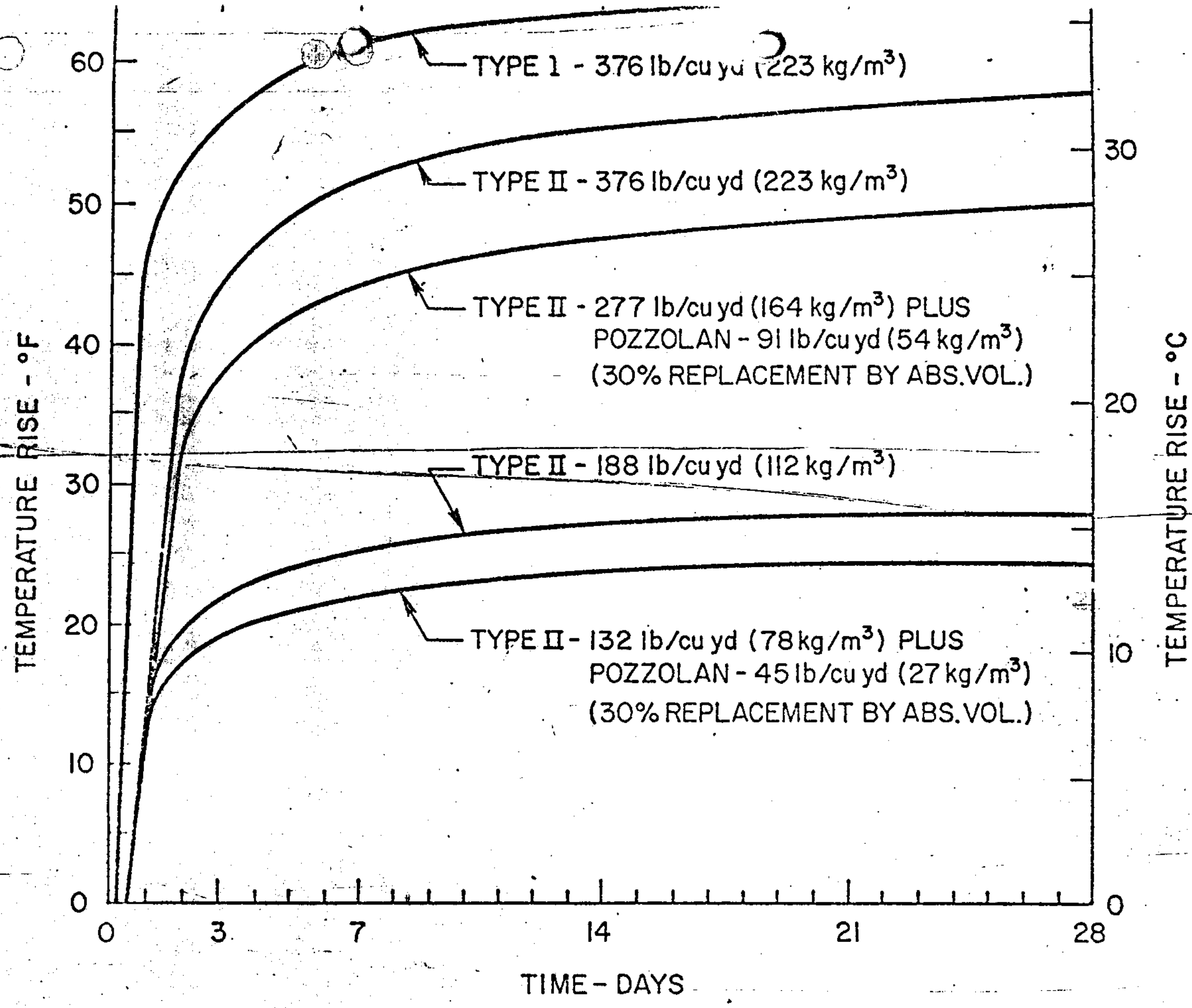
TABLE 2. EFFECT OF TYPE OF AGGREGATE ON DRYING SHRINKAGE OF CONCRETE

Aggregate	1-Year Shrinkage, %
Sandstone	0.097
Basalt	0.068
Granite	0.063
Limestone	0.050
Quartz	0.040

TABLE 3. EFFECT OF MAXIMUM SIZE OF AGGREGATE ON TENSILE STRAIN CAPACITY OF CONCRETE

Mix No.	Aggregate	Maximum Size of Aggregate, in(mm)	W/C+P*	Tensile Strain Capacity, 10^{-6}	
				7 days	28 days
1	Quartzite, natural	3 (75)	0.68	45	71
2	Quartzite, natural	1-1/2 (37.5)	0.68	76	95
3	Quartzite, natural	No. 4 (4.75)	0.68	138	165
4	None (paste)		0.68	310	357
5	Quartzite, crushed	1-1/2 (37.5)	0.68	119	139
6	Quartzite, natural	1-1/2 (37.5)	0.40*	151	145

*All mixes contained 30% fly ash by absolute volume, except No. 6 which had no fly ash.



Original in Luna